

Haoxiang Wan

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Education

University of Houston, Houston, TX

Aug. 2025 -

Ph.D. in Electrical and Computer Engineering

Advisor: [Prof. Xingpeng Li](#), Renewable Power Grid (RPG) Laboratory

Research Focus: Power System Operations and Optimization, Data Center Load Flexibility

University of Illinois at Urbana-Champaign, Urbana, IL

May 2025

Master of Engineering in Energy Systems

The University of Sheffield, Sheffield, UK

June 2024

Bachelor of Engineering in Electrical and Electronic Engineering, *First Class Honours*

Publications

- **H. Wan**, L. Fang, J. Kwon, J. Silva-Rodriguez, J. Yang, J. Yellajosula, F. Jiang, and X. Li, “Impact of Security Modeling Fidelity on Hyperscale Data Center Hosting Capacity Assessment,” submitted to *IEEE PES Grid Edge Technologies Conference & Exposition*, 2027.
- **H. Wan**, J. Kwon, S. Cho, S. B. Sharma, T. Levin, and X. Li, “Risk-Aware Selection of Generation Expansion Portfolios under Uncertain Data-Center Deployment: A Full-Year Cross-Stress Framework,” in preparation for *Applied Energy*, 2026.
- **H. Wan**, J. Kwon and X. Li, “Impact of Data Center Load on Co-Optimized Generation and Transmission Expansion Planning with Embedded Unit Commitment,” submitted to *Hawaii International Conference on System Sciences (HICSS)*, 2027.
- **H. Wan** and X. Li, “Data Center Spatio-Temporal Load Flexibility in Security-Constrained Unit Commitment for Enhanced Grid Efficiency and Reliability,” accepted by *IEEE Industry Applications Society (IAS) Annual Meeting*, 2026. [[arXiv](#)]
- **H. Wan**, L. Fang, and X. Li, “Grid Operational Benefit Analysis of Data Center Spatial Flexibility: Congestion Relief, Renewable Energy Curtailment Reduction, and Cost Saving,” accepted by *IEEE PES General Meeting (PESGM)*, 2026. [[arXiv](#)]
- L. Fang, **H. Wan**, A. Srivastava, and X. Li, “Service-Aware Hosting Capacity Assessment of Edge Data Centers with Workload Flexibility,” *IEEE PES Grid Edge Technologies Conference & Exposition*, 2025.

Research Experience

Graduate Research Assistant

Aug. 2025 – Present

Renewable Power Grid (RPG) Laboratory, University of Houston, Houston, TX

Advisor: [Prof. Xingpeng Li](#)

- Developed Security-Constrained Unit Commitment (SCUC) models with data center spatial and temporal load flexibility, reducing transmission congestion, renewable curtailment, and operating cost.
- Formulated a co-optimized generation and transmission expansion planning (GTEP) model with embedded unit commitment on the IEEE RTS-24 system, quantifying how large data center loads shift investment between transmission and generation.
- Assessed bus-level data center hosting capacity under four security models of increasing fidelity (OPF, B-SCUC, R-SCUC, and N-1 SCUC), isolating the impact of security modeling on interconnection limits.
- Built data pipelines and machine learning models for locational marginal price (LMP) and congestion forecasting in an industry collaboration with Dow.
- Implemented all optimization models in Python/Pyomo, AMPL, and Gurobi.

Undergraduate Capstone: Microcontroller-Based Traffic Signal Control System Sept. 2023 – June 2024

The University of Sheffield, Sheffield, UK

Supervisor: [Dr. Kennedy Ofor](#)

- Designed and implemented an embedded traffic management system programmed in C to optimize urban traffic flow through dynamic signal control.

- Developed a custom four-layer PCB integrating sensor arrays and wireless communication modules.

Teaching Experience

Graduate Teaching Assistant

Jan. 2026 – May 2026

ECE 5380 – Power Electronics & Electric Drives, University of Houston

Instructor: [Dr. Hao Huang](#)

- Supported a synchronous online graduate course covering power semiconductor devices, converters and inverters, DC/induction/synchronous motor drives, harmonics, and filtering.
- Graded homework and examinations for students, providing error-focused feedback to improve learners' comprehension.

Graduate Teaching Assistant

Aug. 2025 – Dec. 2025

ECE 5388 – Renewable Energy Technology, University of Houston

Instructor: [Dr. Hao Huang](#)

- Assisted with instruction and assessment for a course covering solar, wind, and grid integration of renewable resources.
- Provided detailed feedback on assignments and examinations to support student learning outcomes.

Professional Experience

Research Aide (Summer Internship)

May 2026 – Aug. 2026

Argonne National Laboratory, Lemont, IL

Supervisor: [Dr. Jonghwan Kwon](#), Energy Systems and Infrastructure Analysis Division

- Built a Monte Carlo cross-stress framework for generation expansion planning under uncertain data center deployment, testing 30 candidate portfolios against 30 project-level futures on a 254-bus Texas 2030 system.
- Ran 1,161 full-year (8,760-hour) linear programs with GPU solvers (NVIDIA cuOpt/PDLP) in Argonne's [A-LEAF](#) framework, cutting a full-year expansion solve from 24.6 hours on CPU to 1.78 hours while matching CPLEX objectives within 0.5%.
- Identified data center siting mismatch, rather than portfolio capacity, as the dominant driver of undelivered load, and quantified a \$4.12B per year expected-cost saving of scenario-based portfolio selection over planning to the average deployment.
- Conducted hosting capacity analysis and co-optimized generation and transmission expansion planning for hyperscale data center interconnection on the ERCOT network.

Engineering Intern

June 2023 – Aug. 2023

Jiangxi Engineering Research Center of High-Power Electronics and Grid Smart Metering, China

- Contributed to the design of a specialized current detection device for power grid leakage identification, with MATLAB-based simulations validating its reliability under varying operating conditions.

Technical Skills

- **Programming Languages:** Python, Julia, MATLAB, C, C++
- **Optimization & Modeling:** Pyomo, JuMP, AMPL, Gurobi, CPLEX, NVIDIA cuOpt/PDLP, MadNLP
- **Power System Tools:** A-LEAF, MATLAB/Simulink, LTspice
- **Machine Learning:** PyTorch, scikit-learn, pandas, NumPy
- **Other:** LaTeX, Git, Linux/HPC (Slurm, PBS), GPU computing (CUDA)
- **Languages:** English (fluent), Mandarin Chinese (native)

Honors and Awards

- Engineering Excellence Scholarship, The University of Sheffield *2021–2022, 2022–2023*
- NCUK Sheffield Scholarship, The University of Sheffield *2021–2022, 2022–2023*

Professional Affiliations

- Student Member, IEEE Power & Energy Society (PES)
- Student Member, IEEE Industry Applications Society (IAS)